THE BUILDINGS OF GENERALIZED BRAID GROUPS

3.6–bibliography

Let $Y'(b(A))$ denote the component with index $b(A)$. There is an evident projection

$$p' : Y' \longrightarrow Y.$$

Let R be the following relation on Y': for $x \in Y'(b(A))$ and $y \in Y'(b(B))$, $R(x,y)$ means that

$$p'(x) = p'(y) \quad \text{and} \quad p'(x), p'(y) \in Y(A,B).$$

This is an equivalence relation, because, cf. (3.2),

$$Y(A,B) \cap Y(B,C) \subset Y(A,C).$$

One has

$$\text{the } R\text{-saturation of an open subset of } Y' \text{ is again open.} \tag{3.6.1}$$

Put $\widetilde{Y} = Y'/R$. There is an evident projection

$$p : \widetilde{Y} \longrightarrow Y.$$

For every ball $b(A)$ of $\widehat{I}$, let $\widetilde{Y}(b(A))$ denote the image of $Y'(b(A))$ in $\widetilde{Y}$.

The following theorem implies the theorem stated at the beginning of the introduction.

(3.7) Theorem. $\widetilde{Y}$ is a contractible covering of Y.

A. $\widetilde{Y}$ is a covering of Y. We shall show, for $V(F,C)$ as in (3.4)(iv), that $p^{-1}V(F,C)$ is a sum of copies of $V(F,C)$.

a) Let $b(A)$ be a ball of $\widehat{I}$. For B a chamber such that $F \subset B^-$, let C'_B be the chamber of I

$$C'_B = A\, u(A,B)\, u(\pi_F(C), B)^{-1}.$$

By construction, $B \subset q(S(A) \cap S(C'_B))$, and in particular, if $F \neq \{0\}$,

$$[F] \subset q(S(A) \cap S(C'_B)). \tag{3.7.1}$$

We prove that

$$\widetilde{Y}(b(A)) \cap p^{-1}V(F,C) \subset \bigcup_{F \subset B^-} \widetilde{Y}(b(C'_B)). \tag{3.7.2}$$

Let x lie in the left-hand side. Let G be the facet of V to which $\mathcal{R}px$ belongs. One has $F \subset G^-$. Choose B to be a chamber of which G is a facet. By hypothesis,

$$\mathrm{pr}_G\, \mathcal{I}px \in \pi'_G(A) \quad \text{and} \quad \mathrm{pr}_F\, \mathcal{I}px \in C.$$

Thus $\pi'_G(A) = \pi'_G(\pi_F C)$. It follows from (3.2) that every chamber having G as a facet lies in $q(S(A) \cap S(C'_B))$, that px lies in $Y(A, C'_B)$, and hence that $x \in \widetilde{Y}(b(C'_B))$.

b) Put

$$Y'(F,C) = \bigcup_{q(C')=C} \big(Y'(b(C')) \cap p'^{-1}V(F,C)\big).$$

By a), set-theoretically,

$$p^{-1}V(F,C) = Y'(F,C)/R.$$

It is even an isomorphism of topological spaces, by (3.6.1), since $Y'(F,C)$ is open in Y'. By (3.2), the equivalence relation induced by R on

$$\bigcup_{q(C')=C} Y'(b(C'))$$

is trivial. We conclude by observing that, for every C' lying over C,

$$V(F,C) \subset Y(b(C')).$$

B. An open covering of $\widetilde{Y}$.

(3.7.3) Notation. (i) For F a facet of I,

$$\widetilde{Y}(F) = \bigcup_{F\subset S(C)} \big(\widetilde{Y}(b(C)) \cap p^{-1}\mathcal{R}^{-1}\operatorname{Et}(qF)\big).$$

(ii) For $o(A)$, the center of a ball $b(A)$ of $\widehat{I}$,

$$\widetilde{Y}(o(A)) = \widetilde{Y}(b(A)) \cap p^{-1}V(\{0\}, A).$$

(3.7.4) Lemma. (i) The $\widetilde{Y}(s)$, for s a vertex of $\widehat{I}$, form a covering of $\widetilde{Y}$ whose nerve is $\widehat{I}$: a family of $\widetilde{Y}(s)$'s has nonempty intersection if and only if the corresponding s's are the vertices of a facet of $\widehat{I}$.

(ii) If a facet F of I has vertices $s_0, \ldots, s_p$, then

$$\widetilde{Y}(F) = \widetilde{Y}(s_0) \cap \cdots \cap \widetilde{Y}(s_p).$$

(iii) If $F \subset S(A)$, then $\widetilde{Y}(o(A)) \cap \widetilde{Y}(F)$ is contractible; $\widetilde{Y}(o(A))$ is contractible.

(3.7.4.1) Let F_i, $i = 1, 2$, be two facets of I, and C_i, $i = 1, 2$, two chambers of I such that $F_i \subset S(C_i)$ and

$$\big(\widetilde{Y}(b(C_1)) \cap p^{-1}\mathcal{R}^{-1}\operatorname{Et}(qF_1)\big) \cap \big(\widetilde{Y}(b(C_2)) \cap p^{-1}\mathcal{R}^{-1}\operatorname{Et}(qF_2)\big) \neq \varnothing.$$

Then F_1 and F_2 span a facet F of I, unique by (2.5)–(2.11), $F \subset S(C_i)$, and the preceding intersection coincides with

$$\widetilde{Y}(b(C_1)) \cap \widetilde{Y}(b(C_2)) \cap p^{-1}\mathcal{R}^{-1}\operatorname{Et}(F).$$

By hypothesis,

$$Y(C_1, C_2) \cap \mathcal{R}^{-1}\operatorname{Et}(qF_1) \cap \mathcal{R}^{-1}\operatorname{Et}(qF_2) \neq \varnothing;$$

therefore F_1 and F_2 lie in $S(C_1) \cap S(C_2)$ and span a facet F there. Moreover,

$$\operatorname{Et}(qF_1) \cap \operatorname{Et}(qF_2) = \operatorname{Et}(qF),$$

which gives the announced formula.

From (3.7.4.1), it follows that $\widetilde{Y}(F_1) \cap \widetilde{Y}(F_2)$ is empty if F_1 and F_2 do not span a common facet F, and otherwise is equal to

$$\bigcap_{F\subset S(C)} \widetilde{Y}(b(C)) \cap p^{-1}\mathcal{R}^{-1}\operatorname{Et}(F) = \widetilde{Y}(F)$$

(take $C_1 = C_2$). This proves (ii) and part of (i).

(3.7.4.2) If $o(A) \neq o(B)$, then

$$\widetilde{Y}(o(A)) \cap \widetilde{Y}(o(B)) = \varnothing.$$

If $qA = qB$, then $\widetilde{Y}(b(A)) \cap \widetilde{Y}(b(B)) = \varnothing$. If $qA \neq qB$, then

$$p\widetilde{Y}(o(A)) \cap p\widetilde{Y}(o(B)) = \varnothing.$$

(3.7.4.3) If $F \not\subset S(A)$, then

$$\widetilde{Y}(o(A)) \cap \widetilde{Y}(F) = \varnothing.$$

If $F \subset S(B)$, then $F \not\subset S(A) \cap S(B)$ and

$$\operatorname{Et}(qF) \cap Y(A, B) = \operatorname{Et}(qF) \cap p(\widetilde{Y}(b(A)) \cap \widetilde{Y}(b(B))) = \varnothing.$$

It is clear that:

$$\text{if } F \subset S(A), \text{ then } p \text{ induces an isomorphism from } \widetilde{Y}(o(A)) \cap \widetilde{Y}(F) \tag{3.7.4.4}$$

onto

$$Y(qA) \cap \mathcal{R}^{-1}\operatorname{Et}(qF).$$

Lemma (3.7.4) then follows from:

(3.7.5) Lemma. For every facet F of V, and every chamber A,

$$Y(A) \cap \mathcal{R}^{-1}\,\mathrm{Et}(F)$$

is contractible.

Let v_0 be a vector such that $\mathcal{R}v_0 \in F$ and $\mathcal{I}v_0 \in A$. Then, for $0 \le t \le 1$,

$$\varphi_t : v \longmapsto v_0 + t(v - v_0)$$

maps $Y(A)$ and $\mathcal{R}^{-1}\,\mathrm{Et}(F)$ into themselves. One has

$$\varphi_0(Y(A) \cap \mathcal{R}^{-1}\,\mathrm{Et}(F)) = \{v_0\} \quad \text{and} \quad \varphi_1 = \mathrm{Id}\,.$$

C. Study of $\widetilde{Y}(F)$, F a facet of I. By (3.5), (3.7.1), and (3.7.2),

$$\widetilde{Y}(F) = \bigcup_{F \subset B^-} \widetilde{Y}(b(B)) \cap p^{-1}\mathcal{R}^{-1}\,\mathrm{Et}(F). \tag{3.7.6}$$

Choose a fundamental chamber $A_{F,0}$ in $(V_F, \mathcal{M}_F)$, and let I_F be the corresponding building. Also choose $\widetilde{A}_{F,0}$ in I, lying over $\pi_F(A_{F,0})$. For each chamber $B = \widetilde{A}_0.g_F$ of I_F, denote by $\pi_F(B)$ the chamber $\widetilde{A}_{F,0} \cdot \pi_F(g_F)$ of I. By (2.10), (3.7.6) can be rewritten as

$$\widetilde{Y}(F) = \bigcup_{B \text{ in } I_F} \widetilde{Y}(b(\pi_F(B))) \cap p^{-1}\mathcal{R}^{-1}\,\mathrm{Et}(F).$$

Let

$$Y_F = (V_F)_{\mathbb{C}} - \bigcup_{M \in \mathcal{M}_F} M_{\mathbb{C}}$$

and let $\widetilde{Y}_F$ be the covering of Y_F defined using I_F. For any B and C in I_F, it follows from (1.30) that the chambers of S in

$$Y(\pi_F(B), \pi_F(C)) \cap \mathrm{Et}(F)$$

are exactly the $\pi_F(D)$, for D in $Y_F(B, C)$.

The gluings defining $\widetilde{Y}(F)$ and $\widetilde{Y}_F$ are therefore compatible, and there is a unique Cartesian diagram

$$\begin{array}{ccc} \widetilde{Y}_F & \longleftarrow & \widetilde{Y}(F) \\ \downarrow & & \downarrow \\ Y_F & \xleftarrow{\mathrm{pr}_F} & \mathcal{R}^{-1}\,\mathrm{Et}(F) \subset Y \end{array} \tag{3.7.7}$$

which sends

$$\widetilde{Y}(b(\pi_F(B))) \cap p^{-1}\mathcal{R}^{-1}\,\mathrm{Et}(F)$$

to $\widetilde{Y}_F(b(B))$, for B a chamber of I_F. The first vertical arrow, and hence also the second, are coverings.

It is easy to verify that the map pr_F of (3.7.7) is a homotopy equivalence. Hence:

Lemma. $\widetilde{Y}_F$ and $\widetilde{Y}(F)$ have the same homotopy type. (3.7.8)

D. End of the proof. We prove (3.7) by induction on $r = \dim V$. Let $\mathcal{U}$ be the open covering of $\widetilde{Y}$ by the $\widetilde{Y}(s)$, for s a vertex of $\widehat{I}$. By (3.7.4)(i), $\widehat{I}$ is the nerve of $\mathcal{U}$. By (3.7.4), (3.7.8), and the induction hypothesis, the nonempty intersections of opens belonging to $\mathcal{U}$ are contractible. This implies [6] that $\widetilde{Y}$ and $\widehat{I} = \mathrm{Nerf}(\mathcal{U})$ have the same homotopy type. We conclude by (2.15).

4. Braid groups

(4.1) Let V be a finite-dimensional real vector space and $W \subset \mathrm{GL}(V)$ a finite group generated by reflections. Assume $V^W = \{0\}$. Let Φ be any W-invariant Euclidean structure, and let $\mathcal{M}$ be the set of hyperplanes M such that the orthogonal reflection with respect to M lies in W. It is known that:

a) $(V, \mathcal{M})$ satisfies 1.6 ([1], V, 3.9, prop. 7);

b) W permutes the chambers of $(V, \mathcal{M})$ simply transitively ([1], V, 3.2, th. 1);

c) if $x \in V$ belongs to the closed chamber $\overline{A}$, its stabilizer is generated by the reflections with respect to the walls of A containing x ([1], V, 3.3, prop. 1);

d) it follows from c) that W acts freely on

$$Y_W = V_{\mathbb{C}} - \bigcup_{M \in \mathcal{M}} M_{\mathbb{C}}.$$

By b), if C_0 and C_1 are two chambers, there is a unique $w \in W$ such that $wC_0 = C_1$; these w's induce a transitive system of bijections between the walls of the different chambers. Passing to the quotient, one obtains a set D with $\dim(V)$ elements and, for each chamber C, a bijection φ_C from D to the set of walls of C. One has

$$\varphi_{wC}(i) = w\varphi_C(i) \qquad (w \in W). \tag{4.1.1}$$

For the definition of the Coxeter matrix $(m_{ij})_{i,j\in D}$ and of the Coxeter graph, we refer to [1], V, 3.4.

(4.2) Choose a chamber A_0 of $(V, \mathcal{M})$. For $i \in D$, let w_i be the reflection with respect to the wall $\varphi_{A_0}(i)$. [1], V, 3.2, th. 1 says that W is generated by the w_i, with these generators subject only to the relations

$$(w_i w_j)^{m_{ij}} = e.$$

(4.3) Let $X_W = Y_W/W$: by (4.1)d), Y_W is a covering of X_W, with group W. Let $y_0 \in A_0$ have image $x_0 \in X_W$. For each $i \in D$, let $\widetilde{\ell}_i$ be the homotopy class of paths in Y_W, from y_0 to $w_i(y_0)$, which, for $y_0 \in A_0$, contains the broken line with successive vertices

$$y_0, \quad y_0 + iy_0, \quad w_i(y_0) + iy_0, \quad w_i(y_0).$$

The image ℓ_i of $\widetilde{\ell}_i$ in X_W is a loop based at x_0.

(4.4) Theorem. (i) The fundamental group $\pi_1(X_W, x_0)$ is generated by the ℓ_i, subject only to the relations

$$\mathrm{prod}(m_{ij}; \ell_i, \ell_j) = \mathrm{prod}(m_{ji}; \ell_j, \ell_i). \tag{4.4.1}$$

(ii) The universal covering of X_W is contractible: X_W is a $K(\pi, 1)$.

Assertion (i) is proved in Brieskorn [3]. Another proof will be sketched in (4.18) below. As explained in the introduction, (ii) follows from (3.7) and (4.1).

(4.5) In §1 we were largely inspired by Garside's study [4] of Artin's braid group; we can now derive from the results obtained information about all groups of the type $\pi_1(X_W, x_0)$. The translation rests on (4.6) and (4.7) below.

(4.6) Let $G = (C_0, \ldots, C_n)$ be a gallery, and let M_i be the wall separating C_{i-1} from C_i, $1 \le i \le n$. One has

$$\varphi^{-1}_{C_{i-1}}(M_i) = \varphi^{-1}_{C_i}(M_i).$$

Let $\ell(G)$ be the element

$$\varphi^{-1}_{C_1}(M_1) \cdots \varphi^{-1}_{C_n}(M_n)$$

of $L^+(D)$, the free monoid on D. The map ℓ induces a bijection from the set of galleries G with prescribed source to $L^+(D)$. One has

$$\ell(w(G)) = \ell(G) \qquad (w \in W), \tag{4.6.1}$$

$$\ell(G_0G_1) = \ell(G_0)\ell(G_1) \tag{4.6.2}$$

for composable G_0 and G_1. Let

$$w : L(D) \longrightarrow W$$

be the homomorphism extending the map $i \mapsto w_i$ from D to W.

(4.7) Proposition. For every gallery G with source A_0, one has

$$A_0 \cdot G = w(\ell(G))(A_0).$$

This is an avatar of the identity

$$[(w_1 \cdots w_{n-1})w_n(w_1 \cdots w_{n-1})^{-1}][(w_1 \cdots w_{n-2})w_{n-1}(w_1 \cdots w_{n-2})^{-1}] \cdots [w_1] = w_1 \cdots w_n.$$

(4.8) From this proposition it follows that, for $w \in W$, the integer $d(A_0, w(A_0))$ is the minimum length of words in the w_i equal to w. More precisely, ℓ induces a bijection from the set of minimal galleries from A_0 to wA_0 to the set of words $m \in L(D)$, of minimal length, such that $w = w(m)$.

(4.9) Definition. (i) The braid monoid G^+ of type D is the monoid with unit generated by D, whose generators $i \in D$ are subject only to the relations

$$\mathrm{prod}(m_{ij}; i, j) = \mathrm{prod}(m_{ij}; j, i). \tag{4.9.1}$$

(ii) The braid group G, or generalized braid group, of type D, is the group generated by D, with the generators subject to (4.9.1).

(4.10) Notice that W also admits the presentation

$$\mathrm{prod}(m_{ij}; w_i, w_j) = \mathrm{prod}(m_{ij}; w_j, w_i), \tag{4.10.1}$$

$$w_i^2 = e, \tag{4.10.2}$$

so that $i \mapsto w_i$ extends to a homomorphism from G to W.

(4.11) One checks from the definitions that two galleries G_0 and G_1 with the same source are equivalent if and only if $\ell(G_0)$ and $\ell(G_1)$ have the same image in G^+. Taking account of (4.9), (1.12) therefore admits the following translation ([1], IV, 1.6, prop. 5).

(4.12) Reminder. Let $w \in W$. The image in G^+ of the words $m \in L(D)$, of minimal length among those satisfying $w = w(m)$, depends only on w. This image will be denoted $r(w)$; r is a section of $w : G^+ \to W$. By (4.7),

$$r(w) = \ell(u(A_0, wA_0)).$$

Similarly, [1], IV, 1.4, lemma 2 corresponds to (1.11).

(4.13) The quotient category of $\mathrm{Gal}_+(V, \mathcal{M})$ by W is defined as follows:

a) its set of objects is reduced to one element, namely $\mathrm{Ob}(\mathrm{Gal}_+(V, \mathcal{M}))/W$.

b) The monoid of its arrows is $\mathrm{Fl}(\mathrm{Gal}_+(V, \mathcal{M}))/W$, and passage to the quotient

$$\mathrm{Gal}_+(V, \mathcal{M}) \longrightarrow \mathrm{Gal}_+(V, \mathcal{M})/W$$

is a functor.

This category, having only one object, is identified with the monoid of its arrows. By (4.6.1), (4.6.2), and (4.11), ℓ identifies this monoid with G^+.

Similarly one defines $\mathrm{Gal}(V,\mathcal{M})/W$. This category has only one object and, by its universal property, ℓ extends to an isomorphism from the group of its arrows to G:

$$\begin{array}{ccccc} \mathrm{Gal}_+(V,\mathcal{M}) & \longrightarrow & \mathrm{Gal}_+(V,\mathcal{M})/W & \underset{\ell}{\xrightarrow{\sim}} & G^+ \\ \downarrow & & \downarrow & & \downarrow \\ \mathrm{Gal}(V,\mathcal{M}) & \longrightarrow & \mathrm{Gal}(V,\mathcal{M})/W & \underset{\ell}{\xrightarrow{\sim}} & G. \end{array}$$

(4.14) Theorem. (i) In G^+, left and right translations are injective.

(ii) G^+ satisfies Ore's condition on the left and on the right. Thus, by (i), G^+ embeds in G.

(iii) For J a subset of D, let G_J^+, respectively G_J, be the submonoid, respectively subgroup, of G^+, respectively G, generated by the $i \in J$. Then the evident map identifies G_J^+, respectively G_J, with the braid monoid, respectively braid group, of type J.

(iv) One has $G_J^+ = G_J \cap G^+$.

(v) If J and K are two subsets of D, one has $G_J \cap G_K = G_{J\cap K}$.

(vi) Let $n(i)$ be the number of elements of G^+ of length i, and put

$$f = \sum n(i)t^i.$$

For $J \subset D$, let $m(J)$ be the number of walls passing through the intersection of the walls of any chamber A, indexed by J, that is, the number of reflections in the Weyl group W_J. Then

$$f = \left(\sum_{J\subset D} (-1)^{|J|} t^{m(J)}\right)^{-1}.$$

These assertions translate respectively: (i), (1.1)(i) and (ii); (ii), (1.26)(ii) and its transform by passage to opposite galleries, cf. also (4.16) below; (iii), (1.30); (iv), (1.32); (v), (1.31); (vi), (1.21), the f_A of loc. cit. all being equal to f.

Let $x, y \in G^+$. We say that x begins with y if there exists $z \in G^+$ such that $x = yz$. Proposition (1.19)(iii) translates as follows.

(4.15) Lemma. Let $x \in G^+$. There exists a chamber C such that, for every $w \in W$, x begins with $r(w)$ if and only if $wA_0 \in D(A_0, C)$.

(4.16) Let J be a subset of D, and let P be the intersection of the walls $\varphi_{A_0}(i)$, $i \in J$. Let $\Delta(J)$ be the element $\ell(u(A_0, A_0 \cdot A(P)))$ of G^+; put $\Delta = \Delta(D)$. Let $i \mapsto \bar{i}$ be the opposition involution of D. We also denote by $w \mapsto \bar{w}$ the automorphisms of $L^+(D)$, G^+, and G sending i to $\bar{i}$; one has $m_{ij} = m_{\bar{i}\bar{j}}$. Translating (1.26) with the help of (4.4), one finds that, for g in G^+ or in G,

$$g\Delta = \Delta\bar{g}. \tag{4.16.1}$$

In particular, Δ^2 is central.

For every $i \in D$, Δ begins with i: one has $\Delta = ix$ in G^+. Hence:

(4.17) Proposition. G is obtained from G^+ by making the central element Δ^2 invertible.

(4.18) The description (2.11), (2.5)–(2.7) of the building I attached to $(V,\mathcal{M})$ can be translated as follows:

a) the set of vertices of I is
$$I^0 = \coprod_{i\in D} G/G_{D-\{i\}};$$

b) for a set of vertices to span a simplex, it is necessary and sufficient that it be contained in the set of classes $gG_{D-\{i\}}$, for some $g \in G$;

c) the fundamental sphere $S_0 = S(A_0)$ is the union of the fundamental simplex, spanned by the lateral classes of e in the $G/G_{D-\{i\}}$, and its transforms by the $r(w)$, $w \in W$.

The preceding description makes evident a left action of G on I. It respects the set of spheres $S(A)$ and the correspondence $A \mapsto S(A)$.

By transport of structure, G acts on the covering $\widetilde{Y}_W$ of Y_W (3.6), and one has an equivariant morphism

$$(G - \widetilde{Y}_W) \longrightarrow (W - Y_W).$$

It follows that G is the fundamental group of $X_W = Y_W/W$, and (4.4)(i).

(4.19) For completeness, let us show that the word problem, the conjugacy problem, and the question of computing the center of G can be solved as in [4]. If the Coxeter graph D of W is a disjoint union of graphs D_α, the braid group of type D is the product of the braid groups of types D_α. The essential case is therefore that of a connected graph D.

(4.20) The word problem. There is a decision procedure for determining whether two elements m and n of $L(D)$ have the same image in G.

a) Whether m and n in $L^+(D)$ have the same image in G^+ is decidable, because the imposed relations do not change word length and there are only finitely many words of a given length. One may also use (1.22).

b) For any m_i, $i = 1, 2$, in the free group $L(D)$ generated by D, one can compute $m'_i \in L^+(D)$ and $k \geq 0$ such that m_i and $m'_i\Delta^{-k}$ have the same image in G (cf. (4.17)); then m_1 and m_2 have the same image in G if and only if m'_1 and m'_2 have the same image in G^+.

(4.21) Theorem. If the Coxeter graph D is connected, nonempty, and the opposition involution is trivial, respectively nontrivial, the center of G is infinite monogenic, generated by Δ, respectively by Δ^2.

Consider the following property of $x \in G^+$:

$$(*) \qquad \text{for every } i \in D, \text{ there exists } i' \in D \text{ such that } ix = xi'.$$

It is enough to prove that, if x satisfies $(*)$ and $x \neq e$, then $x = \Delta y$ with $y \in G^+$. By induction on the length of x, it will follow that, if x satisfies $(*)$, then x is of the form Δ^k. In particular, the center of G^+ is contained in the set of the Δ^k, $k \geq 0$; by (4.14)(i) and (4.17), the center of G is contained in the set of the Δ^k, $k \in \mathbb{Z}$, and the assertion follows from (4.16.1).

Let $x \in G^+$ satisfy $(*)$. Let C be as in (4.15), and let w be the image in W of $\ell(u(A_0, C))$. We may write $x = r(w)y$, with $y \in G^+$. Let $i \in D$. By $(*)$, $ix = r(w)y'$ begins with $r(w)$; by (1.23), it follows that $ir(w)$ begins with $r(w)$, i.e. $ir(w) = r(w)i''$. In particular, for every i, there exists i'' with $iw = wi''$; this means that every wall of A_0 is also a wall of wA_0. Among the $2^{|D|}$ open simplicial cones bounded by the walls of A_0, only A_0 and $-A_0$ are chambers; they are the only ones whose angles between faces are all $\leq \pi/2$, using here the connectedness of D. If $x \neq e$, then $w \neq e$, and therefore $wA_0 = -A_0$. Since then $\Delta = r(w)$, this completes the proof.

(4.22) The derived group. Let D' be the graph with vertex set D, two vertices being joined by an edge if and only if m_{ij} is odd. Let D_0 be the set of connected components of D'. Define an epimorphism

$$\mathrm{Lg} : G \longrightarrow \mathbb{Z}^{D_0}$$

by sending i to the basis vector indexed by the connected component of i in D'. One verifies at once that Lg identifies $\mathbb{Z}^{D_0}$ with the largest abelian quotient of G.

(4.23) The conjugacy problem. The problem is to give a decision procedure for determining whether two elements x and y of G, given explicitly as images of words in $L(D)$, are conjugate. Since Δ^2 is central, $\Delta^{-2k}a$ and $\Delta^{-2k}b$ are conjugate if and only if a and b are. By (4.17), it is enough to treat the case where $x, y \in G^+$. Similarly, if x and y are conjugate, there is $a \in G^+$ such that

$$xa = ay.$$

If x and y are conjugate, $\mathrm{Lg}(x) = \mathrm{Lg}(y)$. There are only finitely many $x \in G^+$ of given length $\mathrm{Lg}(x)$, and W is finite. The existence of a decision procedure then follows from (4.20) and the following lemma.

(4.24) Lemma. Let $x, y \in G^+$. If x and y are conjugate, there exists a sequence

$$x_0 = x,\ x_1, \ldots, x_n = y$$

of elements of G^+, and a sequence of elements w_i of W, such that

$$x_{i+1} r(w_i) = r(w_i) x_i.$$

Let $a \in G^+$ be such that $xa = ay$. Write

$$a = r(w_0) \cdots r(w_{n-1}),$$

where w_i is the element of W of maximum length such that $r(w_i) \cdots r(w_{n-1})$ can be written in the form $r(w_i) b_i$, with $b_i \in G^+$ (4.15). Let $a_i = r(w_0) \cdots r(w_i)$. We shall prove that

$$x_i = a_i^{-1} x a_i$$

lies in G^+, so that the x_i and w_i answer the problem.

It follows from (1.23) that, for any $w \in W$, if xa begins with $r(w)$, then $xr(w_0)$ also begins with $r(w)$. In particular, since

$$xa = ay = r(w_0) b_0 y$$

begins with $r(w_0)$, one has $xr(w_0) = r(w_0) x_1$ with $x_1 \in G^+$. The proof is completed by induction on n.

Remark. Let σ be an automorphism of the Coxeter graph D, and again denote by σ the corresponding automorphism of G. One has $\Delta^\sigma = \Delta$. The preceding arguments still apply to the question of deciding, for $x, y \in G$, whether there exists $a \in G$ such that $xa = ay^\sigma$. We reduce to taking x, y, and a, in G^+. In that case, with the preceding notation, if

$$xa = ay^\sigma,$$

then the $a_i^{-1} x a_i$ lie in G^+. Taking for σ the involution $i \mapsto \bar{i}$, one obtains a criterion analogous to (4.24) for the conjugacy of Δx and Δy, $x, y \in G^+$.

Bibliography

The reader will find a more complete bibliography on braid groups in [2].

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